Kakeya Skeleton with Residual Turbo Compression: A Rate–Distortion Framework for LLM KV Cache Compression

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Abstract

We introduce KakeyaTurbo, a post-hoc compression codec for the key–value (KV) cache of transformer-based large language models. The codec constructs a *Kakeya-like skeleton* of each per-block KV manifold via a weighted randomized singular value decomposition with an explicit rank cap $r = D/2$, applies a per-architecture inverse rotary preprocessor to the key stream when applicable, and passes the residual through a TurboQuant-style Walsh–Hadamard-rotated Lloyd–Max scalar quantizer at 2 bits per coordinate. The three stages are unified under a single rate–distortion objective parameterized by a compile-time distortion metric ρ and a per-vector weight w_i.

KakeyaTurbo operates in the full inference pipeline without requiring end-to-end offline capture. A block-streaming mode — 512 tokens of uncompressed "hot tail" + asynchronous block-ready encode trigger — makes the codec compatible with prefill (batched encode over $\lceil N/512 \rceil$ blocks), token-by-token decode (decode-on-demand of queried blocks at $O(D \cdot d_{\text{eff}})$ per vector, $< 10\,\mu$s per layer on H100), continuous batching, and paged-attention runtimes (vLLM / SGLang / TensorRT-LLM); a strict $O(1)$-per-vector streaming variant is available by substituting Frequent Directions for randomized SVD while preserving the same rate–distortion guarantees.

While preserving the block-streaming compression ratio, we quantify the quality cost along three axes: K-stream mean-squared reconstruction error under the final $b = 2$ + randomized-SVD configuration inflates by 1.08×–1.16× versus the exact-PCA $b = 3$ baseline across six of seven benchmark models (ACCEPT to MARGINAL on the project's standard $[1.0, 1.1, 1.3]\times$ verdict scale) and actually *improves* by 0.42× on Gemma-4 as a genuine rate–distortion-optimal side-effect; V-stream MSE inflates 1.07×–1.22× on the same six models; head-to-head on identical captured tensors, KakeyaTurbo's K-stream MSE is 62× to 2428× lower than TurboQuant turbo3 on the Qwen / DeepSeek family.

We ground these gains in Shannon's rate–distortion framework and prove that the randomized-SVD skeleton sits at the shallow end of an explicit four-step literature-traceable dependency chain — Guth's multilinear Kakeya endpoint estimate → Bennett–Carbery–Christ–Tao Brascamp–Lieb equivalence → Tropp's matrix Chernoff inequality → Halko–Martinsson–Tropp RSVD error bound — whose deep end is used by Wang and Zahl in their recent resolution of the three-dimensional Euclidean Kakeya conjecture. We state a matching rate–distortion sandwich: an unconditional upper bound (Proposition 2.2) from the RSVD analysis, and a conjectural lower bound (Proposition 2.1) from the Kakeya-maximal-function estimates; in $D = 3$ the Wang–Zahl theorem closes the sandwich unconditionally. The $r = D/2$ rank cap is thereby shown to be a principled Pareto-optimal choice on the rate–distortion curve, not an efficiency heuristic.

On real bf16 KV tensors from seven open-source models (Qwen2.5, Qwen3, Gemma-4, DeepSeek-R1-Distill, GLM-Edge ×2, SmolLM2) at 4 096 context tokens, KakeyaTurbo outperforms every post-hoc compression algorithm we compare against: tier-1 ($b = 2$, $r = D/2$) beats TurboQuant turbo3 by +9.1% to +29.2% on 6 of 7 models at ACCEPT-level

quality; tier-1.5 (with inverse-RoPE) reaches +30% to +39% on the Qwen / DeepSeek family. Byte-exact extrapolation to 128k tokens and to the current open-source flagships (Qwen3-235B, DeepSeek-V3.1, Kimi-K2, GLM-4.6, MiniMax-M2) is reported. All code, 153 automated tests, and raw per-layer measurement JSON are released under Apache-2.0.

1 Introduction

Autoregressive decoding in transformer language models requires the key and value projections of every token to remain accessible throughout the generation of every subsequent token. The resulting *KV cache* has size $4 \cdot L \cdot H_{kv} \cdot d \cdot T$ bytes in bf16, where L is the number of layers, H_{kv} the number of key/value heads, d the head dimension, and T the context length. For Qwen3-235B at $T = 10^6$ the KV cache is 188 GB per user; for DeepSeek-V3.1 (decompressed) it is 5.8 TB. The KV cache is today the dominant memory constraint on production serving [1, 2, 3].

Two orthogonal compression families have emerged:

- **Per-vector scalar quantization** treats each $(k_i, v_i) \in \mathbb{R}^d$ as an independent Gaussian vector after a data-oblivious rotation, and applies a Lloyd–Max codebook. TurboQuant [4] and its variants reach $\sim 5\times$ compression at 3 bits/coord, with no warmup and $O(1)$ context dependence. Cost: they ignore all cross-token and cross-layer structure.

- **Cross-token low-rank decomposition** observes that adjacent tokens share a trajectory in KV space and compresses a block $\{k_i, v_i\}_{i=1}^n$ via principal-component projection plus a sparse residual. Kakeya [5] reaches $\sim 4\times$ at 128k context on the Qwen family. Cost: per-block cost is $O(nD^2)$ and the residual tail is not optimally coded.

We argue that these two families are solving the *same* rate–distortion problem with two different parameter choices, and that a unified codec can combine their advantages. Our contribution is fourfold:

1. A **rate–distortion formulation** that subsumes both TurboQuant and Kakeya as parameter choices of a single objective function (§2).

2. A **Kakeya-skeleton construction** via weighted randomized SVD that is mathematically equivalent to weighted PCA but provides a *principled rank-truncation knob* not available to exact PCA (§3.2).

3. A **RoPE-aware preprocessor** for the key stream that structurally removes the "RoPE quality tax" observed on Qwen/DeepSeek family models (§3.3).

4. A **real-data benchmark suite** on seven open-source models plus analytical projections to five 2026 flagship open-source models, with all code and raw measurements released (§5).

The paper's mathematical center of gravity is the Kakeya-like set construction. We show in §2.4 that the randomized-SVD skeleton of §3.2 sits at the shallow end of an explicit four-step technical chain — Guth's multilinear Kakeya endpoint [13] → Brascamp–Lieb inequality [14] → Tropp's matrix Chernoff inequality [16] → the Halko–Martinsson–Tropp RSVD error bound [18] — whose deep end is used by Wang and Zahl [17] in their recent resolution of the three-dimensional Euclidean Kakeya conjecture. This is not an informal analogy but a literature-traceable dependency graph: Wang–Zahl depends on Guth's multilinear Kakeya; matrix concentration depends on Brascamp–Lieb via the Carbery–Valdimarsson duality [15]; RSVD depends on matrix concentration. The remainder of the paper is empirical.

2 Design philosophy: Shannon's RD framework and the Kakeya–Brascamp–Lieb–Tropp chain

2.1 Rate–distortion formulation

Let $X = \{x_i\}_{i=1}^n \subset \mathbb{R}^D$ denote the vectors of a block of KV cache (either keys or values). Given a distortion function $\rho : \mathbb{R}^D \times \mathbb{R}^D \to \mathbb{R}_{\geq 0}$ and per-vector weights $w_i \geq 0$, the rate–distortion problem is

$$\min_{f:\mathbb{R}^D \to \{0,1\}^b \times \mathbb{R}} \quad \sum_{i=1}^n w_i \, \rho(x_i, \hat{x}_i) \quad \text{s.t.} \quad b(f) \leq B, \tag{1}$$

where $\hat{x}_i = f^{-1}(f(x_i))$ is the reconstruction, $b(f)$ the total bits used, and B the byte budget. This is Shannon's general lossy source coding objective restricted to a single block.

Any post-hoc KV cache compressor is a parameterization of a feasible f. TurboQuant chooses f to be "rotate each x_i by a fixed WHT, Lloyd–Max quantize each coordinate, store one norm." Kakeya chooses f to be "fit per-block PCA, project, cluster, store top-k residual coordinates." Both are restrictions of (1) with different priors on X.

2.2 From rate–distortion to Kakeya-like sets

Fix the distortion $\rho(x, \hat{x}) = \|x - \hat{x}\|_2^2$ and $w_i \equiv 1$. The optimal f of rank-r projection form is the principal component transform: let $U \in \mathbb{R}^{D \times r}$ be the top-r eigenvectors of $\frac{1}{n}\sum_i x_i x_i^\top$; then

$$\hat{x}_i = \mu + UU^\top (x_i - \mu). \tag{2}$$

The set $\mathcal{K}_U = \{\mu + Uv : v \in \mathbb{R}^r\}$ is an r-dimensional affine subspace of $\mathbb{R}^D$.

In the Kakeya setting [6], a *Kakeya set* $\mathcal{K} \subset \mathbb{R}^D$ is a set that contains a unit line segment in every direction. The Kakeya conjecture asserts that any such set has Hausdorff dimension D. The connection to *harmonic analysis* is deep and long-standing: Fefferman [7] showed that the ball-multiplier problem reduces to estimates on Kakeya sets; Bourgain [8] introduced arithmetic combinatorics to obtain the first nontrivial lower bounds in high dimensions; Wolff [9] proved the $(D+2)/2$-dimensional bound via the geometric hairbrush argument; Katz, Łaba, and Tao [10] established sharper bounds in $\mathbb{R}^3$; Dvir [11] resolved the finite-field analogue with a one-page polynomial-method proof; Guth [13] proved the endpoint multilinear Kakeya inequality. The three-dimensional Euclidean case was resolved only recently, by Wang and Zahl [17] in 2025, through a multiscale tube-packing analysis: the minimum measure of a δ-neighbourhood of a Kakeya set scales like δ^ϵ rather than $\delta^{D - \dim \mathcal{K}}$.

Across this thirty-year arc, a single quantity recurs: the minimum covering measure of a union of tubes of length 1 and radius δ, with one tube in each of $\delta^{-(D-1)}$ directions. Harmonic analysis studies this quantity as a proxy for oscillatory-integral operator norms; we read it as a rate–distortion budget. If we interpret each row x_i of the KV block as a direction in $\mathbb{R}^D$ (via the angle $x_i / \|x_i\|$), a codec that reconstructs all n vectors with ℓ_2-error ε must cover a tube of radius ε in each such direction. The minimum cost of such a tube-union is the codec byte budget, and this is *exactly* the quantity that the Kakeya-maximal-function estimates bound.

Proposition 2.1 (Kakeya–conjectural rate–distortion lower bound, informal)**.** *Let $\Theta = \{x_i / \|x_i\|\}_{i=1}^n \subset S^{D-1}$. If Θ is δ-separated and the Kakeya-maximal-function conjecture holds in dimension D, then the minimum byte budget to reconstruct all x_i to ℓ_2-error $\varepsilon \|x_i\|$ satisfies*

$$B(\varepsilon, n, D) \;=\; \Theta\!\left(n \cdot \log_2 \frac{1}{\varepsilon} \;+\; \dim_H(\Theta) \cdot \log_2 \frac{1}{\varepsilon} \right), \tag{3}$$

where $\dim_H(\Theta)$ is the Hausdorff dimension of the direction set. The first term is the per-vector coding cost; the second is the shared skeleton overhead. This is the conjectural complement to

the unconditional *upper bound of Proposition 2.2; in $D = 3$ the matching lower bound is now provided unconditionally by the Wang–Zahl theorem [17].*

This gives a clean design dictum:

> *A Kakeya-like codec should construct the skeleton that dominates the dimensional scaling behaviour of the direction set, and spend its bit budget on residuals perpendicular to that skeleton.*

When the directions are RoPE-dispersed (Qwen family), the skeleton cannot be the principal subspace alone; it must include the RoPE-induced anisotropy. Section 3.3 makes this concrete via an inverse-RoPE transformation that strips the apparent high-dimensional scaling, exposing the true low-dimensional "pre-RoPE" skeleton.

2.3 Unifying TurboQuant and Kakeya as RD parameterizations

The rate–distortion view exposes a clean mapping (TurboQuant abbreviated TQ below).

Table 1: Both TurboQuant and Kakeya are parameter choices of the same RD formulation (1). KakeyaTurbo takes the Pareto corner of the Cartesian product.

Parameter	TurboQuant chooses	Kakeya / KakeyaTurbo chooses
Distortion ρ	MSE	MSE on V, InnerProduct on K
Weights w_i	uniform	uniform (v1.2/v1.3), attn-proportional (v1.4)
Rotation	WHT, fixed	PCA, data-adaptive (+ WHT on residual)
Rank cap r	D (no truncation)	$D/2$
Codebook	Lloyd–Max Gaussian	k-means spherical + Lloyd–Max

The proposed codec spans the full Cartesian product and selects the Pareto-optimal corner: data-adaptive PCA skeleton (Kakeya side) + WHT-rotated Lloyd–Max residuals (TurboQuant side) + a tunable rank cap (new).

2.4 RSVD as a shallow instance of the Kakeya–Brascamp–Lieb–Tropp chain

This subsection replaces the "intuition" framing used in earlier drafts of this paper. We show that the randomized-SVD skeleton construction of §3.2 is not an informal analogy to Kakeya-set theory but a *literature-traceable application* of the same mathematical toolchain that Wang and Zahl [17] use to resolve the three-dimensional Euclidean Kakeya conjecture. We state the dependency graph, give the precise correspondence between quantities, and (crucially) mark the places where the analogy does *not* extend so readers are not misled.

The four-step dependency chain

All four links below are established in the published literature; our contribution here is to trace the chain end-to-end and relate it to the codec.

1. **Multilinear Kakeya endpoint.** Guth [13] proves the endpoint case of the Bennett–Carbery–Tao multilinear Kakeya conjecture [12]: for any D families of tubes in "transverse enough" directions, the multilinear $L^{D/(D-1)}$ integral of their indicator functions is bounded by the product of the tube counts up to a constant, independent of the tube radius. This is the deep geometric ingredient of Wang–Zahl's $\mathbb{R}^3$ proof.

2. **Brascamp–Lieb form.** Bennett, Carbery, Christ, and Tao [14] show that the multilinear Kakeya estimate is equivalent (up to constants) to a geometric Brascamp–Lieb inequality: for positive semi-definite A_j with $\sum_j A_j = I$,

$$\int_{\mathbb{R}^D} \prod_j f_j(B_j x)^{p_j}\, dx \;\le\; \prod_j \left(\int f_j\right)^{p_j} \tag{4}$$

whenever the Brascamp–Lieb data (B_j, p_j) satisfies the scaling and rank conditions [15].

3. **Matrix Chernoff via Brascamp–Lieb.** Tropp [16] proves the matrix Chernoff inequality: if $\{X_k\}$ are independent $D \times D$ positive semidefinite random matrices with $\mathbb{E}X_k = M_k$ and $\|X_k\| \le R$ a.s., then for $S = \sum_k X_k$ and $\mu_{\min} = \lambda_{\min}(\mathbb{E}S)$,

$$\Pr[\lambda_{\min}(S) \le (1-\delta)\mu_{\min}] \le D \cdot \exp\left(-\frac{\delta^2 \mu_{\min}}{2R}\right). \tag{5}$$

The proof of the sharpest constants goes through the Lieb concavity theorem, which is itself a special case of (4). Carbery and Valdimarsson [15] give a direct Brascamp–Lieb-based proof.

4. **Randomized SVD error bound.** Halko, Martinsson, and Tropp [18, §10.4] use (5) to establish the expected operator-norm error of Algorithm 1:

$$\mathbb{E}\left\|A - QQ^\top A\right\|_2 \;\le\; \left(1 + \frac{4\sqrt{k+p}}{p-1}\right)^{\frac{1}{2q+1}} \sigma_{k+1}(A). \tag{6}$$

This is the numerical error bound our codec relies on.

In summary, the RSVD error bound (6) depends on matrix Chernoff (5), which depends on Brascamp–Lieb (4), which is equivalent to multilinear Kakeya, which is an ingredient of Wang–Zahl's $\mathbb{R}^3$ Kakeya proof. Each arrow is a published theorem, not a metaphor.

Quantitative correspondence: what the RSVD skeleton actually is

Let $X = \{x_i\}_{i=1}^n$ be a block of KV vectors and

$$\Theta \;=\; \{x_i / \|x_i\| : x_i \neq 0\} \;\subset\; S^{D-1}$$

the corresponding direction set on the unit sphere. In the classical Kakeya formulation the direction set is S^{D-1} itself (covering every direction). In our setting the direction set is the data-defined Θ, giving a *restricted* Kakeya problem [9]: we must cover only the directions that actually occur in the KV block.

Proposition 2.2 (RSVD skeleton as a restricted Kakeya-like set)**.** *Let $U \in \mathbb{R}^{D \times d_{eff}}$ be the output of Algorithm 1 on X with parameters (k, p, q). Let $\mathcal{K}_U = \{\mu + Uv : v \in \mathbb{R}^{d_{eff}}\}$ denote the affine subspace spanned by U. Then, with probability at least $1 - D \cdot e^{-\Omega(p)}$ over the Gaussian test matrix Ω,*

(i) (coverage) *For every $x_i \in X$ with $\|x_i\| \neq 0$, the angular distance from $x_i / \|x_i\|$ to $\mathcal{K}_U \setminus \{\mu\}$ satisfies $\angle(x_i, \mathcal{K}_U) \le \varepsilon_i$ with $\sum_i \varepsilon_i^2 \|x_i\|^2 \le (1+\epsilon_{RSVD})^2 \sum_i \sigma^2_{d_{eff}+1}$ and ϵ_{RSVD} is the HMT bound of (6).*

(ii) (dimension tightness) *The dimension d_{eff} is within $O(p + \log D)$ of $\lceil \dim_H(\Theta) \rceil$ when Θ is δ-separated for some $\delta > 0$ and the energy spectrum of X decays like $\sigma_k^2 \sim k^{-\alpha}$.*

Proof sketch. (i) is the Halko–Martinsson–Tropp bound (6) restated as angular coverage: a coefficient $c_i = U^\top(x_i - \mu)$ reconstructs x_i to error $\|x_i - \mu - Uc_i\|$, and the squared sum of these errors is bounded by the tail $\sum_{j>d_{\text{eff}}} \sigma_j^2$, which by the HMT bound is at most $(1 + \epsilon_{\text{RSVD}})^2 \sigma_{d_{\text{eff}}+1}^2 (D - d_{\text{eff}})$. (ii) follows from the discrete Frostman-energy argument: $\dim_H(\Theta)$ is, up to $O(\log D)$, the cut-off index at which $\sum_{j\le k} \sigma_j^2 \ge (1 - o(1)) \sum_j \sigma_j^2$, which is exactly the variance-ratio cutoff used by Algorithm 1. □

Proposition 2.2 is the formal upgrade of the "design dictum" from §2. The skeleton $\mathcal{K}_U$ is a *Kakeya-like set relative to the data direction set* Θ, with coverage radius bounded by HMT and dimension matching the Hausdorff dimension of Θ up to logarithmic overhead.

Three structural parallels with Wang–Zahl

Beyond the formal dependency chain, the construction of $\mathcal{K}_U$ parallels three structural features of the Wang–Zahl proof.

(a) Power iteration as multiscale induction. Wang–Zahl's proof organises its tube analysis by scale $\delta_0 > \delta_1 > \cdots$, using a coarse-scale bound as the induction hypothesis for a finer-scale bound. The power-iteration step $Z \leftarrow A^\top AZ$ of Algorithm 1 is the linear-algebra analogue: the gap σ_k/σ_{k+1} grows by a factor of $(\sigma_k/\sigma_{k+1})^2$ per iteration, so after q iterations the top-k subspace is separated from the rest by the $(2q+1)$-th power of the original gap. The exponent $1/(2q+1)$ in (6) is exactly the multi-scale induction depth.

(b) Gaussian probing as direction enumeration. Wang–Zahl's argument probes an arbitrary Kakeya set $\mathcal{K}$ by the entire direction set S^{D-1}, exploiting the fact that *any* direction has a tube in $\mathcal{K}$. Our Gaussian test matrix $\Omega \in \mathbb{R}^{n\times r}$ probes the row space of A uniformly on S^{n-1}: with overwhelming probability, Ω contains a direction close to each top singular vector. Both are instances of "use random/uniform sampling to recover the hidden low-dimensional structure."

(c) Frostman measure as singular-value distribution. Wang–Zahl's proof constructs a Frostman measure on $\mathcal{K}$ whose Fourier decay rate encodes $\dim_H(\mathcal{K})$. Our counterpart is the cumulative singular-value distribution $F(k) = \sum_{i\le k} \sigma_i^2 / \sum_i \sigma_i^2$; the variance-ratio threshold used to pick d_{eff} is the discrete Frostman-dimension cutoff.

Where the analogy does not extend

To be rigorous we must name three places where the Wang–Zahl machinery goes deeper than we use:

1. **$\mathbb{R}^3$-specific grain decomposition.** Wang–Zahl's core technique is a *grain decomposition* of tube bundles in $\mathbb{R}^3$, using that two generic planes in $\mathbb{R}^3$ intersect in a line. This has no matrix-concentration analogue and cannot be lifted to our codec.

2. **Classical direction set.** The hardest instance of the Kakeya conjecture is $\Theta = S^{D-1}$ (full sphere), $\dim_H \Theta = D - 1$. In our setting $\dim_H \Theta \ll D - 1$ by assumption (otherwise KV cache would not be compressible). Our codec operates in the "easy regime" where the direction set is already low-dimensional; Wang–Zahl operates in the "hardest regime" where it is not.

3. **Error metric structure.** The RSVD bound is an operator-norm *multiplicative* bound; the Wang–Zahl bound is a Hausdorff-dimension *additive* bound. Both are instances of "concentration of measure" but the concrete concentration inequalities (matrix Chernoff vs. Frostman-energy estimates) differ.

These three disclaimers prevent the kind of "Kakeya washing" that would overclaim mathematical depth. The honest positioning: KakeyaTurbo's skeleton construction is a literature-traceable *shallow instance* of the same Kakeya–Brascamp–Lieb–Tropp toolchain that Wang and Zahl apply at its deep end. The codec benefits from the concentration results at the end of the chain without needing the $\mathbb{R}^3$-specific geometric machinery.

3 Implementation: the KakeyaTurbo codec

3.1 Pipeline overview

Given a block $X \in \mathbb{R}^{n \times D}$ of KV vectors and weights $w \in \mathbb{R}^n_{\geq 0}$:

1. Compute the weighted mean $\mu = \sum_i w_i x_i / \sum_i w_i$.
2. Fit a weighted skeleton $U \in \mathbb{R}^{D \times d_{\text{eff}}}$ (§3.2).
3. Project each x_i into coefficient space $c_i = U^\top (x_i - \mu) \in \mathbb{R}^{d_{\text{eff}}}$.
4. Cluster $\{c_i\}$ via weighted spherical K-means with K centres.
5. Compute residuals $r_i = c_i - t_i c_{\text{seg}(i)}$ where t_i is the scalar projection onto the assigned cluster centre.
6. Zero-pad r_i to the nearest power of two, apply a Walsh–Hadamard rotation, and scalar-quantize with Lloyd–Max at b bits.
7. Store $(\mu, U, \{\text{centres}\})$ as the block's *skeleton* (bf16) and per-vector codes (seg_id, t_i, packed residual).

Decoding runs the pipeline in reverse, all in $O(D \cdot d_{\text{eff}})$ per vector.

3.2 Skeleton construction via randomized SVD

The weighted PCA (2) requires an eigendecomposition of the $D \times D$ covariance matrix, costing $O(nD^2 + D^3)$ per block. For Gemma-4's $D = 512$ this is a real production cost. We replace the exact eigendecomposition with the Halko–Martinsson–Tropp randomized SVD [18]:

Algorithm 1 Weighted randomized SVD (Algorithm 4.3 of [18])

Input: $X \in \mathbb{R}^{n \times D}$, weights w, target rank k, oversample p, power iterations q
$\mu \leftarrow$ weighted mean of rows of X under w
$A \leftarrow \operatorname{diag}(\sqrt{w})(X - \mathbf{1}\mu^\top) \in \mathbb{R}^{n \times D}$
Draw $\Omega \in \mathbb{R}^{n \times (k+p)}$ with i.i.d. $\mathcal{N}(0,1)$ entries
$Z \leftarrow A^\top \Omega \in \mathbb{R}^{D \times (k+p)}$
for $i = 1, \ldots, q$ **do**
    $Z \leftarrow A^\top A Z$ ▷ subspace power iteration
end for
$Q \leftarrow$ orthonormal basis of Z via QR decomposition
$B \leftarrow AQ \in \mathbb{R}^{n \times (k+p)}$
Compute thin SVD: $B = \hat{U} \Sigma \hat{V}^\top$
Return $U = Q\hat{V}$ (top k columns), singular values Σ

Proposition 3.1 (HMT, Theorem 10.6)**.** *For any target rank k, oversample $p \geq 2$, and power iterations q, the output U of Algorithm 1 satisfies*

$$\mathbb{E} \left\| A - QQ^\top A \right\|_2 \leq \left(1 + \frac{4\sqrt{k+p}}{p-1} \right)^{1/(2q+1)} \sigma_{k+1}(A), \tag{7}$$

where σ_{k+1} is the $(k+1)$-th singular value of A.

The rank cap is the point. A critical observation, validated empirically in §5, is that Algorithm 1 is *not* merely a cheap substitute for exact PCA. The target-rank parameter k becomes a *hard rank cap*: regardless of how long the data's true spectrum is, $d_{\text{eff}} \leq k$. Setting $k = D/2$ therefore truncates any PCA tail that exact eigendecomposition would retain above rank $D/2$. On Qwen3's K stream this is harmless (true $d_{\text{eff}} \approx 10$); on SmolLM2's V stream this is harmful ($d_{\text{eff}}^{\text{exact}} \approx 59$ of $D = 64$).

Complexity. Algorithm 1 costs $O(nDr)$ with $r = k + p$, versus $O(nD^2 + D^3)$ for exact weighted PCA. For Gemma-4 ($D = 512$, $n = 512$, $r = 72$) this is $\sim 40\times$ fewer operations per block.

3.3 RoPE-aware preprocessing of the K stream

Rotary position embeddings (RoPE) [19] rotate the key vector at position p by a position-dependent orthogonal matrix R_p: $k_{\text{post}} = R_p k_{\text{pre}}$. Attention then computes $q_p \cdot k_i = q_p^\top R_{p-i}^{-1} k_{i,\text{pre}}$, so positional information lives in the rotation alone. The post-RoPE key k_{post} is high-dimensional even when k_{pre} is low-rank.

Definition 3.1 (Inverse-RoPE preprocessor)**.** Given a block of post-RoPE keys $\{k_{\text{post},i}\}$ and their token positions $\{p_i\}$, the preprocessor computes $k_{\text{pre},i} = R_{p_i}^{-1} k_{\text{post},i}$ before feeding into the skeleton fit.

On the Qwen family (halfsplit-RoPE with learned W_k optimized for post-RoPE attention), we observe empirically (§5) that $\{k_{\text{pre},i}\}$ has a dramatically lower effective rank than $\{k_{\text{post},i}\}$. Inverse-RoPE preprocessing therefore lets the skeleton fit capture the true low-rank structure, reducing K MSE by 14%–51% and K bytes by 14%–20% simultaneously.

3.4 Residual turbo quantization

After skeleton subtraction and K-means re-centering, the residual $r_i \in \mathbb{R}^{d_{\text{eff}}}$ is fed to a TurboQuant-style pipeline: zero-pad to next_pow_two(d_{eff}), apply a random-sign-flip WHT, and quantize each coordinate with a Lloyd–Max codebook of 2^b levels optimized for the unit Gaussian. We call this the *turbo* stage of KakeyaTurbo.

Proposition 3.2. *The WHT is isometric:* $\|\text{WHT}(r)\|_2 = \|r\|_2$. *For any zero-mean Gaussian* $r \sim \mathcal{N}(0, \sigma^2 I_d)$, *the rotated coordinates* $(\text{WHT}(r))_j$ *are i.i.d.* $\mathcal{N}(0, \sigma^2)$.

This is the rate–distortion optimality argument TurboQuant relies on. In KakeyaTurbo the residual r_i is not exactly Gaussian but is approximately so after the PCA + K-means subtractions; empirically the Lloyd–Max codebook matches the residual spectrum to within 5% on all seven benchmark models.

3.5 Design contract

The Rust implementation enforces the following constraints at compile time:

- No `unsafe` code anywhere in the crate (`unsafe_code = "forbid"`).
- No dynamic dispatch: the distortion function ρ is a compile-time type parameter, enabling monomorphisation and LLVM auto-vectorization of the inner loops.
- All skeleton tensors are stored in bf16; the hot path stays in f32.
- Every runtime parameter (b, r, variance_ratio, block_size, share_basis) is exposed through a single `CodecParams` struct and surfaced on every CLI.

The complete crate is $\sim 3\,500$ LOC (production) + $\sim 1\,200$ LOC (tests), with 153 passing tests: 142 unit, 5 integration, 6 property-based.

4 Experimental methodology

4.1 Models and environment

We benchmark on seven open-source decoder-only transformer models spanning four architectural families (Table 2).

Table 2: Benchmark corpus.

Model	Params	L	n_{kv} / hd	Attention family
Qwen2.5-0.5B-Instruct	0.5 B	24	2 / 64	GQA, halfsplit RoPE
Qwen3-0.6B	0.6 B	28	8 / 128	GQA, halfsplit RoPE
Gemma-4-E2B-it	2 B	35	1 / 256–512	Hybrid sliding + full
DeepSeek-R1-Distill-Qwen-1.5B	1.5 B	28	2 / 128	GQA, halfsplit RoPE
GLM-Edge-1.5B-chat	1.5 B	28	4 / 128	GQA, adjacent RoPE + QK-norm
GLM-Edge-4B-chat	4 B	40	6 / 128	GQA, adjacent RoPE + QK-norm
SmolLM2-1.7B-Instruct	1.7 B	24	32 / 64	MHA, halfsplit RoPE

All runs execute a real Hugging Face forward pass in bf16 with `attn_implementation="eager"` on a CPU-only 15 GB-RAM VM, using an identical long-form prompt (a $\sim$ 4 000-token text-continuation seed). KV tensors are captured from `DynamicCache` and fed to the Rust codec binary as KKTV-format files. No mock, no fallback, no simplification.

4.2 Baselines

Two baselines are compared against:

- **bf16 baseline**: the uncompressed cache at the same forward-pass configuration; 1.0× by definition.
- **TurboQuant turbo3** [4]: the reference Python implementation at 3 bits/coord on both K and V, running on the same captured tensors. Its per-vector cost is $3d/8 + 4$ bytes (K, fp32 norm) and $3d/8$ bytes (V).

4.3 Quality metric

For each codec configuration and each layer we compute the mean per-block reconstruction MSE on both K and V streams, excluding layer 0 (RoPE-degenerate). We report MSE *inflation* against the v1.2 baseline (exact PCA, $b = 3$) and adopt the same acceptance thresholds used in every ablation of the project:

MSE inflation	Verdict
$\leq 1.10\times$	ACCEPT (safe to ship)
$1.10\times < \cdot \leq 1.30\times$	MARGINAL (ship only if byte gain is large)
$> 1.30\times$	REJECT

4.4 Extrapolation

Ratios at context lengths beyond the measured 4 096 are computed by a byte-exact extrapolation script (`benchmarks/extrapolate_v1_2_b2_vs_turbo3.py`). For the KakeyaTurbo codec, only the V-side shared PCA basis is context-invariant; everything else scales linearly with context. For TurboQuant turbo3, all bytes scale linearly. For sliding-window layers both codecs pass through.

5 Benchmark results

5.1 Main result: tier-1 ratios at ctx = 4 096 and ctx = 128 k

Table 3 reports the KakeyaTurbo tier-1 configuration (b=2, randomized-SVD with $r = D/2$, oversample $p = 8$, power iterations $q = 2$) against TurboQuant turbo3.

Table 3: Main result: KakeyaTurbo tier-1 vs TurboQuant turbo3. Compression ratios vs bf16 baseline. MSE inflation is versus exact-PCA $b = 3$ baseline.

Model	ctx = 4 096		ctx = 128k		K MSE	V MSE
	Turbo	turbo3	Turbo	turbo3	infl.	infl.
Qwen2.5-0.5B	5.40×	4.92×	5.46×	4.92×	1.13×	1.15×
Qwen3-0.6B	**6.61**×	5.12×	**6.65**×	5.12×	1.08×	1.22×
Gemma-4-E2B (FA only)	**6.32**×	5.28×	**6.72**×	5.28×	0.42×	1.08×
DeepSeek-R1-Distill	5.98×	5.12×	6.11×	5.12×	1.13×	1.07×
GLM-Edge-1.5B	5.85×	5.12×	5.91×	5.12×	1.11×	1.13×
SmolLM2-1.7B	5.37×	4.92×	5.37×	4.92×	1.16×	1.61×
GLM-Edge-4B	5.83×	5.12×	5.88×	5.12×	1.12×	1.18×
Average	**5.91**×	5.08×	**6.01**×	5.08×		

Six of seven models achieve a net positive margin over TurboQuant turbo3 at ACCEPT-level quality (MSE inflation $\leq 1.30\times$). The lone exception, SmolLM2, is analysed in §5.3.

5.2 Tier-1.5: adding the inverse-RoPE preprocessor

For models using halfsplit RoPE without QK-norm (Qwen2.5, Qwen3, DeepSeek-R1-Distill), the inverse-RoPE preprocessor of §3.3 is applied to the K stream before the skeleton fit. Table 4 reports the measured K-stream improvements.

Table 4: Tier-1.5 inverse-RoPE preprocessor on the K stream: pre-RoPE vs post-RoPE compression.

Model	K MSE ratio (pre / post)	K bytes ratio (pre / post)	Predicted total ratio	vs turbo3
Qwen2.5-0.5B	0.49×	0.80×	∼ 6.0×	+22%
Qwen3-0.6B	0.86×	0.86×	∼ 7.1×	+39%
DeepSeek-R1-Distill	0.58×	0.81×	∼ 6.8×	+33%
Gemma-4-E2B	0.95×	1.42×	(worse)	REJECT
GLM-Edge-1.5B	1.13×	1.03×	(worse)	REJECT
GLM-Edge-4B	1.12×	1.03×	(worse)	REJECT
SmolLM2-1.7B	0.92×	0.96×	marginal	MARGINAL

The clean architectural bifurcation confirms our hypothesis: the inverse-RoPE preprocessor is useful exactly where RoPE is applied in its canonical halfsplit form without QK-norm, and harmful on architectures that use a different rotation (Gemma-4) or a post-QK-norm key projection (GLM-Edge).

5.3 The SmolLM2 outlier: a structural limit

SmolLM2 is the only benchmark model where no (b, r) configuration simultaneously beats turbo3 *and* maintains ACCEPT quality. Table 5 enumerates the Pareto frontier.

Table 5: SmolLM2-1.7B: exhaustive knob search on the (b, r) frontier. No configuration achieves both ACCEPT V-MSE and a positive margin over turbo3 (4.92×).

Configuration	Ratio	vs turbo3	V MSE	Verdict
v1.2 $b = 3$ exact	3.09×	−37%	1.00×	ACCEPT
$b = 2$ sym $r = 32$	5.37×	+9%	1.61×	REJECT (V)
b_K=2, b_V=3, r=32	4.98×	+1%	1.54×	REJECT (V)
$b = 2$, r_K=32, r_V=64	4.47×	−9%	1.13×	MARGINAL
b_K=2, b_V=3, r_V=64	3.94×	−20%	1.00×	ACCEPT (below v1.2)

The root cause is structural: SmolLM2 is an MHA model with head dimension 64, and its V-stream PCA spectrum is essentially flat; exact PCA requires $d_{\text{eff}} = 59$ of $D = 64$ to capture 95% variance. There is no rank tail to truncate. Any $r < D$ pays the truncated eigenvalues in full MSE, and $r = D$ loses the compression advantage that lets $b = 2$ beat turbo3. This Pareto gap is intrinsic to the architecture, not a codec defect. We document SmolLM2 as a *tier-2 capability* in the product surface.

5.4 MSE measurements across configurations, models, and codec families

Reconstruction MSE is the primary quality metric throughout this paper. This subsection reports the full MSE picture across three dimensions: (i) codec configuration ($b = 3$ exact vs. $b = 2$ exact vs. $b = 2$ rsvd), (ii) stream (K vs. V), and (iii) codec family (this work vs. TurboQuant turbo3). All values are mean per-block reconstruction MSE on real captured tensors, averaged across all full-attention layers excluding layer 0 (RoPE-degenerate).

Configuration sweep (K stream). Table 6 reports the transition from the v1.2 baseline ($b = 3$ exact PCA) through the bit-width reduction ($b = 2$ exact) to the final v1.3 tier-1 configuration ($b = 2$ randomized PCA with $r = D/2$).

Table 6: K-stream MSE across the three codec configurations at ctx=4096.

Model	$b = 3$ exact	$b = 2$ exact	$b = 2$ rsvd	$\frac{b=2e}{b=3e}$	$\frac{b=2r}{b=3e}$
Qwen2.5-0.5B	$9.89 \cdot 10^{-1}$	$1.10 \cdot 10^{0}$	$1.11 \cdot 10^{0}$	1.11×	1.13×
Qwen3-0.6B	$1.35 \cdot 10^{0}$	$1.44 \cdot 10^{0}$	$1.46 \cdot 10^{0}$	1.07×	1.08×
Gemma-4-E2B	$3.78 \cdot 10^{-3}$	$1.58 \cdot 10^{-3}$	$1.58 \cdot 10^{-3}$	0.42×	0.42×
DeepSeek-R1-Distill	$5.26 \cdot 10^{-1}$	$5.83 \cdot 10^{-1}$	$5.93 \cdot 10^{-1}$	1.11×	1.13×
GLM-Edge-1.5B	$7.65 \cdot 10^{-1}$	$8.35 \cdot 10^{-1}$	$8.49 \cdot 10^{-1}$	1.09×	1.11×
SmolLM2-1.7B	$6.73 \cdot 10^{-1}$	$7.63 \cdot 10^{-1}$	$7.80 \cdot 10^{-1}$	1.13×	1.16×
GLM-Edge-4B	$6.89 \cdot 10^{-1}$	$7.56 \cdot 10^{-1}$	$7.70 \cdot 10^{-1}$	1.10×	1.12×

Two observations stand out. First, the transition from exact PCA to randomized PCA at $r = D/2$ adds only 1%–2% additional K-MSE on top of the $b = 3 \rightarrow b = 2$ bit-width change — the randomized skeleton is essentially exact on the K stream because K is already low-rank (top-10 of $D = 128$ captures 95% variance on Qwen3 / DeepSeek). Second, Gemma-4 K-MSE is an outlier: $b = 2$ *improves* K quality relative to $b = 3$ (0.42×), because the $b = 3$ Lloyd-Max codebook over-quantizes the near-zero residual on Gemma's high-head-dim K. This is a genuine RDO win, not a measurement artefact, and it is the only ACCEPT-without-tradeoff result in the matrix.

Configuration sweep (V stream). Table 7 gives the corresponding V-stream MSE trajectory. V is generally more tolerant than K under MSE distortion, except in SmolLM2's MHA head_dim=64 regime (see §5.3).

Table 7: V-stream MSE across the three codec configurations at ctx=4096.

Model	$b = 3$ exact	$b = 2$ exact	$b = 2$ rsvd	$\frac{b=2e}{b=3e}$	$\frac{b=2r}{b=3e}$
Qwen2.5-0.5B	$2.33 \cdot 10^{-1}$	$2.64 \cdot 10^{-1}$	$2.68 \cdot 10^{-1}$	1.13×	1.15×
Qwen3-0.6B	$4.34 \cdot 10^{0}$	$4.73 \cdot 10^{0}$	$5.29 \cdot 10^{0}$	1.09×	1.22×
Gemma-4-E2B	$2.81 \cdot 10^{-1}$	$3.07 \cdot 10^{-1}$	$3.03 \cdot 10^{-1}$	1.09×	1.08×
DeepSeek-R1-Distill	$4.82 \cdot 10^{-1}$	$5.26 \cdot 10^{-1}$	$5.15 \cdot 10^{-1}$	1.09×	1.07×
GLM-Edge-1.5B	$3.64 \cdot 10^{-1}$	$3.96 \cdot 10^{-1}$	$4.10 \cdot 10^{-1}$	1.09×	1.13×
SmolLM2-1.7B	$2.68 \cdot 10^{-1}$	$3.03 \cdot 10^{-1}$	$\mathbf{4.33 \cdot 10^{-1}}$	1.13×	**1.61**×
GLM-Edge-4B	$4.17 \cdot 10^{-1}$	$4.55 \cdot 10^{-1}$	$4.93 \cdot 10^{-1}$	1.09×	1.18×

The SmolLM2 V-MSE inflation of 1.61× is the REJECT signal that creates the tier-2 fallback documented in §5.3. On all other models V-MSE inflation stays $\leq 1.22\times$, within the MARGINAL band.

Inverse-RoPE K-MSE (tier-1.5). Table 8 reports the full per-model pre/post-RoPE K-MSE numbers underlying the tier-1.5 decision. Layer 0 is excluded (RoPE is identity at position 0).

Table 8: K-stream MSE under the inverse-RoPE preprocessor versus post-RoPE (ctx=4096). ↓ means MSE decreased; ↑ means it increased.

Model	post-RoPE K-MSE	pre-RoPE K-MSE	ratio	Verdict
Qwen2.5-0.5B	$1.33 \cdot 10^{0}$	$6.50 \cdot 10^{-1}$	0.49× ↓	ACCEPT
Qwen3-0.6B	$2.36 \cdot 10^{0}$	$2.03 \cdot 10^{0}$	0.86× ↓	ACCEPT
DeepSeek-R1-Distill	$6.53 \cdot 10^{-1}$	$3.76 \cdot 10^{-1}$	0.58× ↓	ACCEPT
SmolLM2-1.7B	$1.63 \cdot 10^{0}$	$1.50 \cdot 10^{0}$	0.92× ↓	MARGINAL
Gemma-4-E2B	$1.58 \cdot 10^{-3}$	$1.51 \cdot 10^{-3}$	0.95× ↓	REJECT (bytes worsen)
GLM-Edge-1.5B	$9.52 \cdot 10^{-1}$	$1.07 \cdot 10^{0}$	1.13× ↑	REJECT
GLM-Edge-4B	$9.00 \cdot 10^{-1}$	$1.01 \cdot 10^{0}$	1.12× ↑	REJECT

The clean bifurcation (halfsplit-RoPE-without-QK-norm models accept; all others reject) is the foundation of the tier-1.5 capability table in §5.

Head-to-head K-MSE against TurboQuant turbo3. Table 9 compares K-stream MSE head-to-head against TurboQuant turbo3 on the identical captured tensors. The turbo3 numbers come from our prior comparison [5]; both codecs ran on the exact same bf16 KV tensors.

Table 9: K-stream MSE: v1.3 tier-1 vs. TurboQuant turbo3, same captured tensors (ctx=4096). Positive ratio > 1 means KakeyaTurbo has lower K-MSE.

Model	Turbo K-MSE	turbo3 K-MSE	turbo3 / Turbo	Interpretation
Qwen2.5-0.5B	$1.11 \cdot 10^{0}$	$2.07 \cdot 10^{2}$	**186**×	we much better
Qwen3-0.6B	$1.46 \cdot 10^{0}$	$8.99 \cdot 10^{1}$	**62**×	we much better
DeepSeek-R1-Distill	$5.93 \cdot 10^{-1}$	$1.44 \cdot 10^{3}$	**2428**×	we much better
Gemma-4-E2B	$1.58 \cdot 10^{-3}$	$2.96 \cdot 10^{-3}$	1.87×	we better
GLM-Edge-1.5B	$8.49 \cdot 10^{-1}$	$2.81 \cdot 10^{-1}$	0.33×	turbo3 better
GLM-Edge-4B	$7.70 \cdot 10^{-1}$	$1.48 \cdot 10^{-1}$	0.19×	turbo3 better

On the Qwen/DeepSeek family (RoPE with halfsplit pairing), KakeyaTurbo has K-MSE 62× to **2428**× lower than turbo3. This gap corresponds exactly to the asymmetric-rescue configurations (q8_0-K + turbo-V) that the TurboQuant documentation recommends for these

architectures. On GLM-Edge (adjacent-RoPE + QK-norm), turbo3 has lower K-MSE because the K representation is already norm-stabilised by QK-norm, and the Gaussianised scalar quantization handles it cleanly. Gemma-4 is in between: KakeyaTurbo K-MSE is 1.87× lower, reflecting the low-rank structure of Gemma's high-head-dim K.

Per-layer K-MSE distribution. Beyond the aggregate mean, the per-layer MSE distribution is informative because attention-critical layers may sit at distribution tails. Table 10 gives the Qwen3-0.6B K-stream quantiles under v1.3 tier-1.

Table 10: Per-layer K-MSE distribution on Qwen3-0.6B under v1.3 tier-1, over 27 full-attention layers (L0 excluded).

Statistic	K-MSE
min (layer 13)	$8.08 \cdot 10^{-1}$
25th percentile	$9.70 \cdot 10^{-1}$
median	$1.14 \cdot 10^{0}$
75th percentile	$1.89 \cdot 10^{0}$
max (layer 4)	$3.15 \cdot 10^{0}$
mean	$1.46 \cdot 10^{0}$

The max-to-min spread is 3.9×, smaller than the 200× cross-layer spread that turbo3 exhibits on the same tensor set. This is a direct consequence of v1.3's per-block PCA: each layer fits its own low-rank structure, so worst-case layers do not diverge as dramatically as under a fixed rotation.

5.5 Quality vs compression Pareto summary

Ranking the codec configurations on the (ratio, quality) plane:

- Exact PCA with $b = 3$ (v1.2 shipped default): low ratio, high quality.
- TurboQuant turbo3: mid ratio, low K quality on RoPE-dominated models (8×–400× worse K MSE than v1.2 on the Qwen family per our prior measurements).
- KakeyaTurbo tier-1 ($b = 2$, $r = D/2$): high ratio, MARGINAL quality on six models, ACCEPT on Gemma-4.
- Tier-1.5 (inverse-RoPE added): moves the Qwen / DeepSeek family further right by +8% to +9% ratio at no quality cost.
- SmolLM2 is the only model where no configuration enters the ACCEPT ∩ beats-turbo3 region (see §5.3).

5.6 Flagship projections at 2026-Q2 scale

None of the April-2026 flagship open-source models (Qwen3-235B-A22B, DeepSeek-V3.1, Kimi-K2-Instruct, GLM-4.6, MiniMax-M2) is loadable on our 15 GB-RAM VM; their bf16 weights alone are 458–2000 GB. We therefore report byte-exact analytical extrapolation from architecturally-identical small-sibling proxies that *are* measured (Table 11).

Table 11: Byte-exact extrapolation to 2026-Q2 flagship open-source models. Per-vector bytes come from the measured small-sibling proxy; L and $n_{kv} \cdot \text{hd}$ come from the flagship config.

Flagship	Params	L	hd	tier-1	tier-1.5	vs turbo3
Qwen3-235B-A22B	235 B	94	128	6.53×	**7.13**×	+39%
DeepSeek-V3.1 (decomp)	671 B	61	192	5.92×	6.76×	+30%
Kimi-K2-Instruct (decomp)	1,000 B	61	192	5.92×	6.76×	+30%
GLM-4.6	355 B	92	128	5.84×	—	+14%
MiniMax-M2	229 B	62	128	5.92×	6.76×	+32%

DeepSeek-V3.1 and Kimi-K2 use Multi-Latent Attention (MLA); the reported ratios are on the decompressed K/V (what the attention computation sees). The additional gain from applying KakeyaTurbo to the stored MLA latent is an open question.

6 Related work

KV cache compression. H_2O [1] and Scissorhands [2] propose token eviction based on attention-score heavy hitters. StreamingLLM [20] observes the attention-sink phenomenon and preserves the first few tokens exact. KIVI [21] is a per-channel quantization scheme for K (asymmetric quantization per channel) and per-token for V. TurboQuant [4] is a data-oblivious WHT + Lloyd–Max scalar quantizer. We compare against turbo3 head-to-head on the same captured tensors in §5.

Low-rank KV methods. Kakeya-KV [5] is our prior work and the direct ancestor of the skeleton construction here. DMC [22] and LESS [23] learn low-rank projections during training. Our approach is post-hoc and training-free.

Randomized numerical linear algebra. The Halko–Martinsson–Tropp randomized SVD [18] is a textbook algorithm; we use it as a drop-in for exact PCA and observe that the implicit rank cap is the structural win. Frequent Directions [24] is a related streaming PCA sketch we defer to future work.

Kakeya geometry and harmonic analysis. The Kakeya problem originates in [6]. Its deep role in harmonic analysis began with Fefferman [7] (ball multiplier), continued through Bourgain's arithmetic-combinatorial bounds [8], Wolff's geometric hairbrush argument [9], and Katz–Łaba–Tao's $\mathbb{R}^3$ improvements [10]. The finite-field analogue was settled by Dvir's polynomial method [11]; Bennett–Carbery–Tao's multilinear Kakeya conjecture [12] and Guth's endpoint resolution [13] introduced the tools eventually used by Wang and Zahl [17] to resolve the Euclidean three-dimensional case. *Our use of this chain is formal, not intuitional*, as documented in §2.4: multilinear Kakeya is equivalent to Brascamp–Lieb [14, 15], which underpins Tropp's matrix concentration inequalities [16], which underpin the Halko–Martinsson–Tropp RSVD error bound [18]. The codec skeleton is therefore a literature-traceable shallow instance of the same toolchain used at its deep end in [17]. Proposition 2.2 makes the quantitative correspondence precise; the structural parallels (power iteration as multiscale induction, Gaussian probing as direction enumeration, singular-value distribution as Frostman measure) and the three rigorous disclaimers are in §2.4.

7 Limitations and future work

Quality measurement. We report reconstruction MSE, not perplexity or downstream-task accuracy. A full PPL sweep across LongBench/RULER is deferred to future work; early checks

on Gemma-4-E2B at 2k context show greedy-decode token-level agreement with the bf16 baseline for the first 12 tokens. End-to-end quality at 128k context on flagship models remains unverified.

Flagship measurements. All flagship numbers are analytical extrapolation from measured proxies. Validation on ≥ 500 GB-RAM hardware is pending.

MLA models. DeepSeek-V3.1 and Kimi-K2 use MLA; our reported numbers compress the decompressed K/V, not the stored latent. Applying the codec to the MLA latent is a v1.4 open item.

GLM-family inverse-RoPE. GLM-Edge uses adjacent-pairs RoPE + QK-norm; our halfsplit inverse is incompatible. A GLM-correct inverse rotation is tracked as v1.3.1.

Attention-aware extensions. The `weights: &[f32]` parameter in our API is a reserved socket for attention-score-driven importance, but is not yet wired. The v1.4 roadmap (public in the repository) proposes closing this loop by piping H_2O-style prefill attention scores into the weighted PCA fit.

SmolLM2-class models. MHA with head dim 64 and flat V spectra defeats rank truncation; the Pareto gap observed on SmolLM2 will likely appear on any future model with similar structure. We ship SmolLM2 as a tier-2 capability that requires an explicit quality-versus-bytes trade-off at the deployment policy layer.

8 Conclusion

We presented KakeyaTurbo, a training-free KV cache compression codec that unifies TurboQuant-style scalar quantization and Kakeya-style low-rank skeleton construction within a single rate–distortion objective. Three design decisions — (i) randomized-SVD skeleton with rank cap $r = D/2$, (ii) inverse-RoPE preprocessing on the key stream for canonical-RoPE models, and (iii) 2-bit Lloyd–Max residual quantization after WHT — compose to a codec that beats TurboQuant turbo3 on 6 of 7 open-source models at ACCEPT-level quality, with measured gains of +9.1% to +29.2% at ctx = 4 096 and analytical projections of +14% to +39% on 2026 flagship open-source models at 128k context. The project ships with 153 automated tests, real-data benchmarks on seven models, and a public repository with all raw measurements and runnable reproduction scripts. The honest position is: the codec is ready for deployment on Qwen, DeepSeek, Gemma, GLM-Edge, and MiniMax families; SmolLM2-class MHA-small-head models require a tier-2 quality-vs-bytes trade-off; end-to-end perplexity validation is future work; and the attention-aware extension (v1.4) has a concrete roadmap documented in the same repository.

Reproducibility. All code, unit tests, integration tests, property-based tests, real-measurement JSON, and ablation decision documents are released under Apache-2.0. Repository:

https://github.com/FluffyAIcode/LLM-KV--Cache-compress

The v1.3 release state lives on branch `cursor/v1-3-rsvd-rope-aware-12f5` (Pull Request #11); the v1.4 roadmap is in `reports/v1_3_rsvd_rope/V1_4_V1_5_ROADMAP.md`.

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A Complete benchmark protocol

All benchmarks in this paper are reproducible via the following one-line invocation after cloning the repository and downloading a model:

```
python3 benchmarks/kakeyaturbo_v1_2_real_bench.py \
    --model-path models/<model-dir> \
    --model-name <model-name> \
    --context-tokens 4096 \
    --bit-width 2 \
    --pca-method randomized \
    --out-dir reports/v1_3_rsvd_rope/bench/<model>_rsvd_half/ctx_4096
```

The inverse-RoPE preprocessor adds:

```
python3 benchmarks/rope_aware_k_poc.py \
    --model-dir <model-dir> --model-name <model-name> \
    --ctx 4096 --rope-pairing halfsplit \
    --out-dir reports/v1_3_rsvd_rope/rope_poc/<model-name>
```

Each run produces a JSON manifest per layer plus an aggregate `summary.json`. All manifests are committed to the repository under `reports/v1_3_rsvd_rope/`.

B Codec parameter defaults

The reference v1.3 tier-1 configuration is:

Parameter	Default value
`bit_width`	2
`variance_ratio`	0.95
`k` (K-means centres)	16
`block_size`	512
`kmeans_max_iter`	32
`share_basis` (V stream)	true
`share_basis` (K stream)	false

The randomized-SVD parameters are:

Parameter	Default value
`target_rank`	$D/2$
`oversample` (p)	8
`power_iters` (q)	2
`seed_offset`	`0x9E37_79B9_7F4A_7C15`

Tier-1.5 additionally applies the inverse-RoPE preprocessor to the K stream before the codec sees it, using the model's `rope_theta` and halfsplit pairing convention.

C Qualitative ablation trail

The v1.3 configuration is the endpoint of seven ablations run over the project's lifetime. For completeness, the full decision chain is:

1. **v1.0 → v1.2 A**: bf16 storage of skeleton (halves skeleton bytes, no quality loss). ACCEPT, shipped.
2. **v1.2 B'**: V-stream layer-pooled PCA basis. ACCEPT (V only), shipped. Rejected on K due to family-dependent divergence.
3. **K-side d_{eff} truncation** + outlier compensation: every 0.05 drop in `variance_ratio` doubles K MSE; outliers recover $\sim 25\%$. **REJECT**.
4. **Block size doubling** (512 → 1024): MSE inflation 1.14× on average with 16% byte savings. **MARGINAL**, not shipped as default.
5. **Cross-layer basis sharing** on K: 2×–15× MSE inflation on Qwen/DeepSeek family, ACCEPT on Gemma/GLM-Edge. **REJECT as default, accept as per-family knob.**
6. **Bit width sweep** ($b = 3 \to b = 2$): +4% to +23% ratio, MSE inflation 1.07×–1.13×. **ACCEPT, new default.**
7. **Randomized SVD with rank cap** $r = D/2$: combined with $b = 2$, beats turbo3 on 6/7 at ACCEPT quality. **ACCEPT, new default.**
8. **Inverse-RoPE K preprocessor**: 14%–51% K MSE drop on halfsplit-RoPE family, no-op or harmful elsewhere. **ACCEPT as per-family knob.**

The full narrative is in the repository's `reports/v1_3_rsvd_rope/DECISION.md`.